



Engenharia Informática

Gestão de Sistemas e de Redes

- Ano Letivo 2021/2022 -

Level 2 Step 1 Normal

Reading, Writing and no Arithmetic

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Índice

1. O que representa o termo alinhamento de partição? É relevante quando usamos um disco SSD?	3
4. Crie uma VM Debian Linux com 5 discos virtuais de 100 MB cada, mais o disco do sistema. Descreva as etapas necessárias para realizar a seguinte sequência de tarefas no comando linha: Descreva as etapas necessárias (incluindo a instalação de pacotes adicionais) para executar a seguinte sequência de tarefas na linha de comando	3
5. Pesquise e descreva.....	10

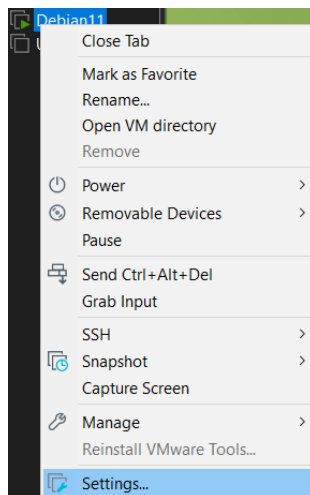
1. O que representa o termo alinhamento de partição? É relevante quando usamos um disco SSD?

Alinhamento de partição representa o alinhamento entre clusters, caso não estejam alinhados pode resultar uma degradação de desempenho no disco. Dado isto, os clusters devem estar alinhados em 512bytes (1 bloco).

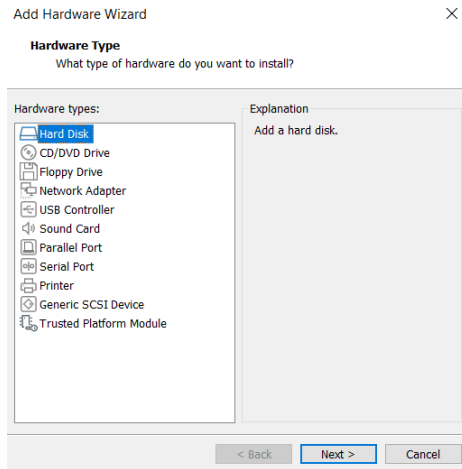
Nos discos SSD, devido ao número limitado de operações, os dados deverão estar alinhados quando não acontece, o SSD é obrigado a reescrever toda a informação que o bloco continha, o que faz com que a performance do SSD diminua.

4. Crie uma VM Debian Linux com 5 discos virtuais de 100 MB cada, mais o disco do sistema. Descreva as etapas necessárias para realizar a seguinte sequência de tarefas no comando linha: Descreva as etapas necessárias (incluindo a instalação de pacotes adicionais) para executar a seguinte sequência de tarefas na linha de comando:

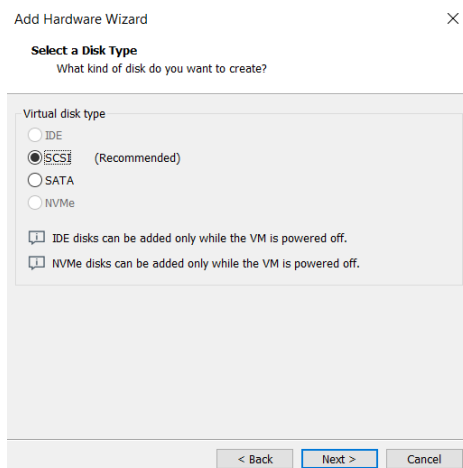
Para adicionar um disco novo carregamos com o botão direito na nossa VM e selecionamos a opção “Settings...”



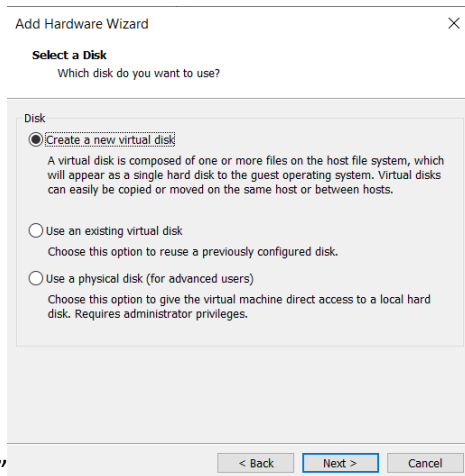
Com a opção “Hard Disk” selecionada,
fazemos “Next”



Escolhemos a opção recomendada,
“SCSI”

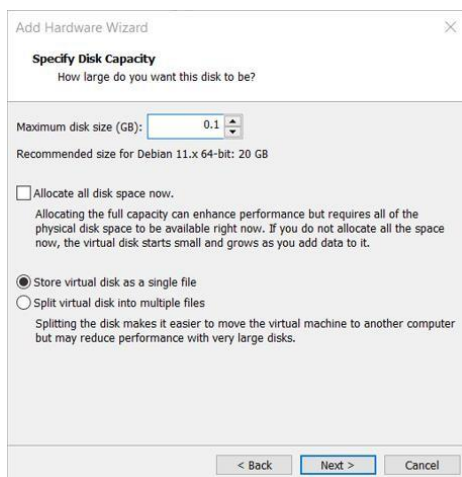


Para, então, criar um disco novo, selecionamos a opção “Create a new virtual

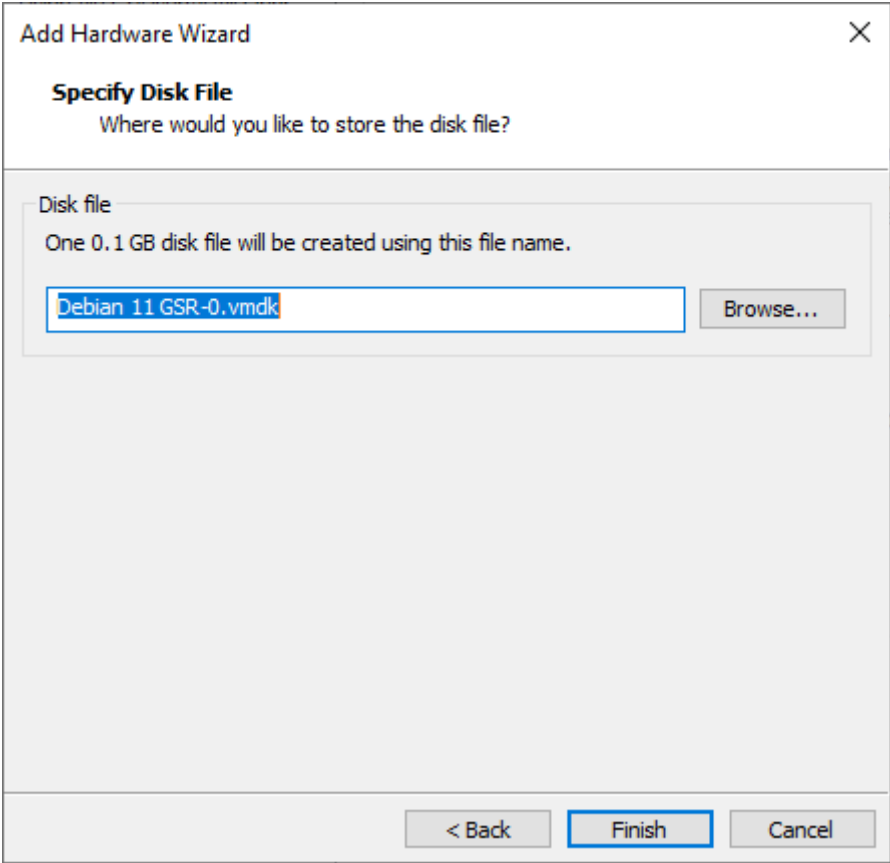


disk”

Especificamos as informações que nos são dadas no exercício

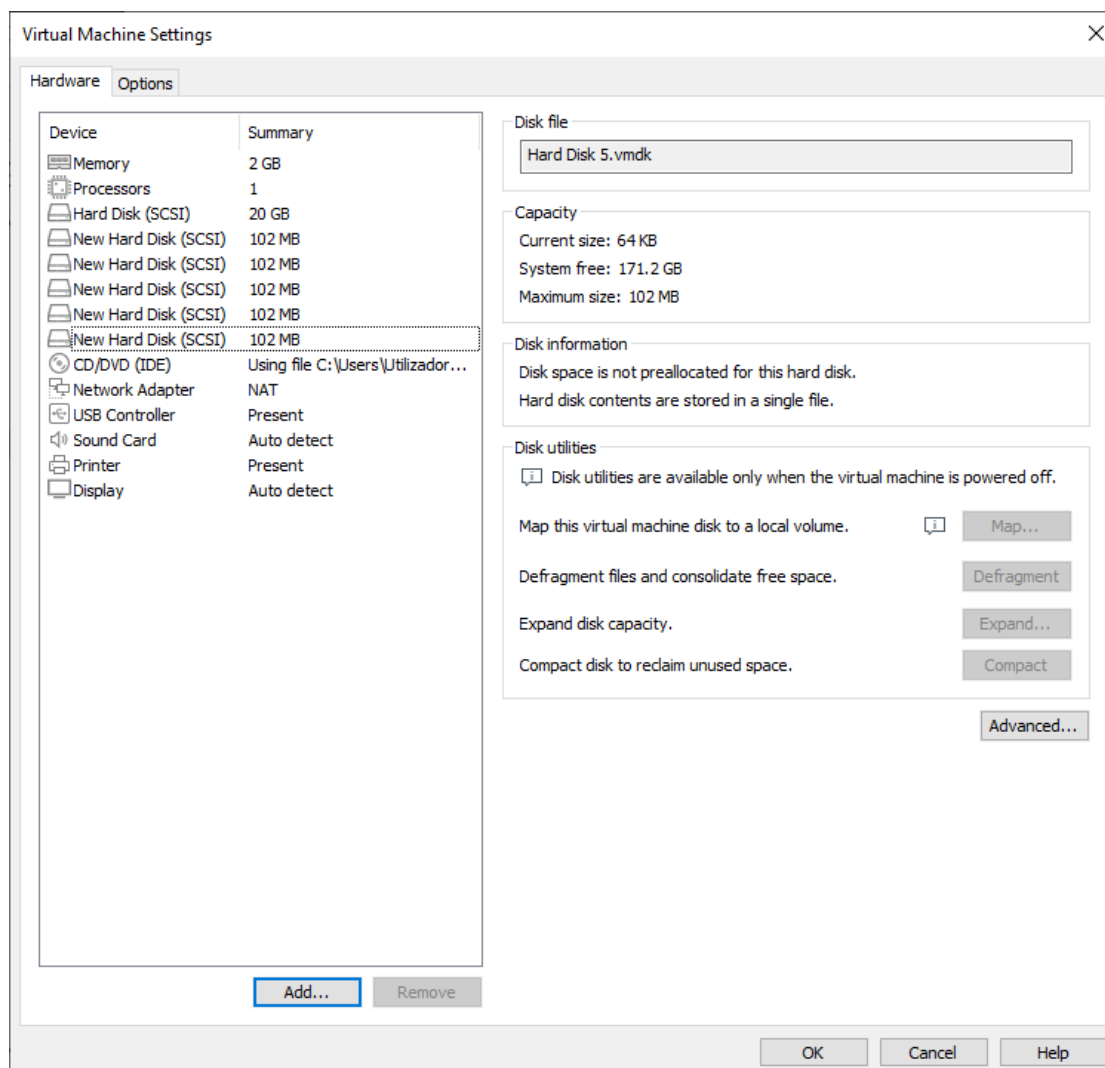


Nomear o disco.



The image shows a Windows-style dialog box titled "Add Hardware Wizard" with a close button (X) in the top right corner. The main heading is "Specify Disk File" followed by the question "Where would you like to store the disk file?". Below this, there is a section labeled "Disk file" containing the text "One 0.1 GB disk file will be created using this file name." and a text input field with the value "Debian 11 GSR-0.vmdk". To the right of the input field is a "Browse..." button. At the bottom of the dialog, there are three buttons: "< Back", "Finish" (which is highlighted with a blue border), and "Cancel".

Após adicionar os 5 discos, a janela deverá aparecer desta forma:



- i. Usando o mdadm, organize os primeiros 4 discos em uma matriz RAID 5 com 3 ativos e 1 sobressalente. Nomeie seu dispositivo / dev / md0.

Para instalarmos o mdadm devemos recorrer ao comando:

```
sudo apt-get install mdadm -y
```

Para mostrar os discos que o sistema possui usamos o seguinte comando: `lsblk`

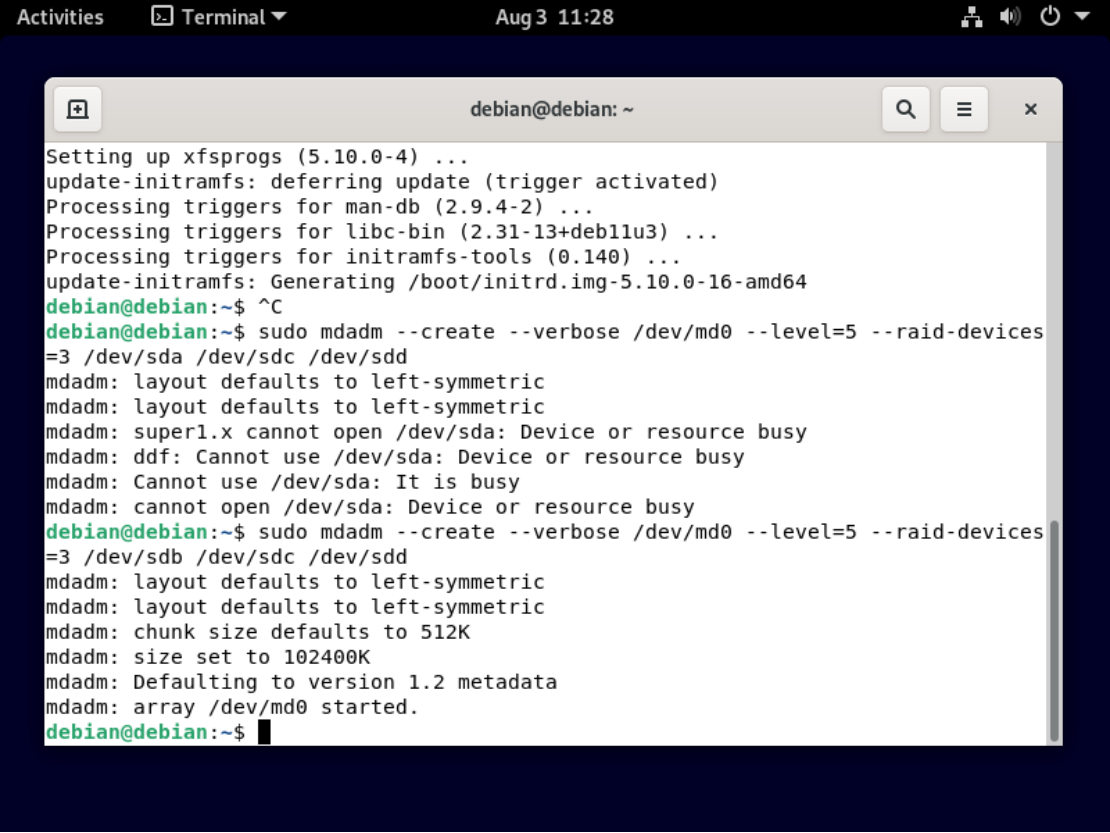
```
debian_jl@debian: ~  
debian_jl@debian:~$ lsblk  
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT  
sda          8:0    0   20G  0 disk  
├─sda1       8:1    0   19G  0 part /  
├─sda2       8:2    0    1K  0 part  
└─sda5       8:5    0   975M  0 part [SWAP]  
sdb          8:16   0    2G  0 disk  
sdc          8:32   0   102M  0 disk  
sdd          8:48   0   102M  0 disk  
sde          8:64   0   102M  0 disk  
sdf          8:80   0   102M  0 disk  
sdg          8:96   0   102M  0 disk  
sr0         11:0    1   378M  0 rom  
debian_jl@debian:~$
```

Para criar o RAID tipo 5 com 3 discos ativos, usamos o comando:

```
sudo mdadm --create --verbose /dev/md0 --level=5 --raid-devices=3 /dev/sdc /dev/sdd /dev/sde
```

Para adicionar o host-space ao RAID, usamos o comando:

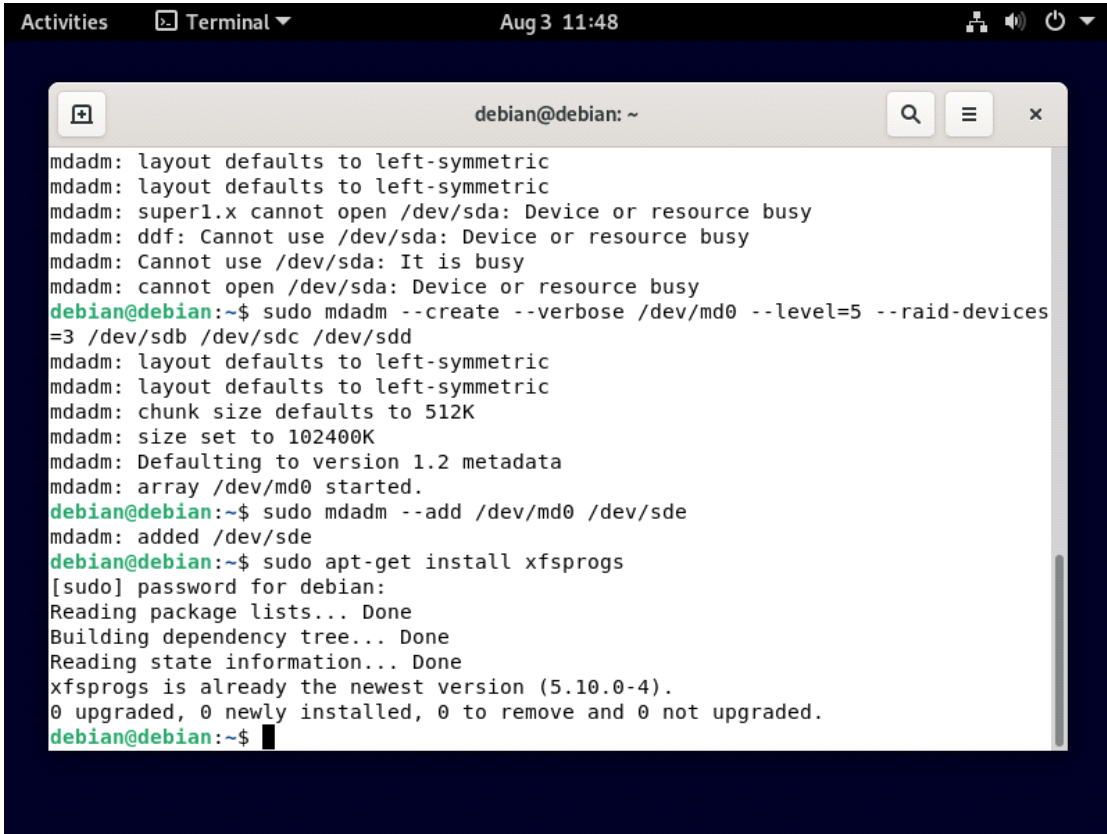
```
sudo mdadm --add /dev/md0 /dev/s
```



```
Activities Terminal Aug 3 11:28
debian@debian: ~
Setting up xfsprogs (5.10.0-4) ...
update-initramfs: deferring update (trigger activated)
Processing triggers for man-db (2.9.4-2) ...
Processing triggers for libc-bin (2.31-13+deb11u3) ...
Processing triggers for initramfs-tools (0.140) ...
update-initramfs: Generating /boot/initrd.img-5.10.0-16-amd64
debian@debian:~$ ^C
debian@debian:~$ sudo mdadm --create --verbose /dev/md0 --level=5 --raid-devices=3 /dev/sda /dev/sdc /dev/sdd
mdadm: layout defaults to left-symmetric
mdadm: layout defaults to left-symmetric
mdadm: super1.x cannot open /dev/sda: Device or resource busy
mdadm: ddf: Cannot use /dev/sda: Device or resource busy
mdadm: Cannot use /dev/sda: It is busy
mdadm: cannot open /dev/sda: Device or resource busy
debian@debian:~$ sudo mdadm --create --verbose /dev/md0 --level=5 --raid-devices=3 /dev/sdb /dev/sdc /dev/sdd
mdadm: layout defaults to left-symmetric
mdadm: layout defaults to left-symmetric
mdadm: chunk size defaults to 512K
mdadm: size set to 102400K
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
debian@debian:~$
```

- ii. Crie um sistema de arquivos XFS na matriz e disponibilize -o em /mnt / md0. Certifique-se de que o volume esteja montado no momento da inicialização.

Usamos o comando: `sudo apt-get install xfsprogs` para instalar o XFS.

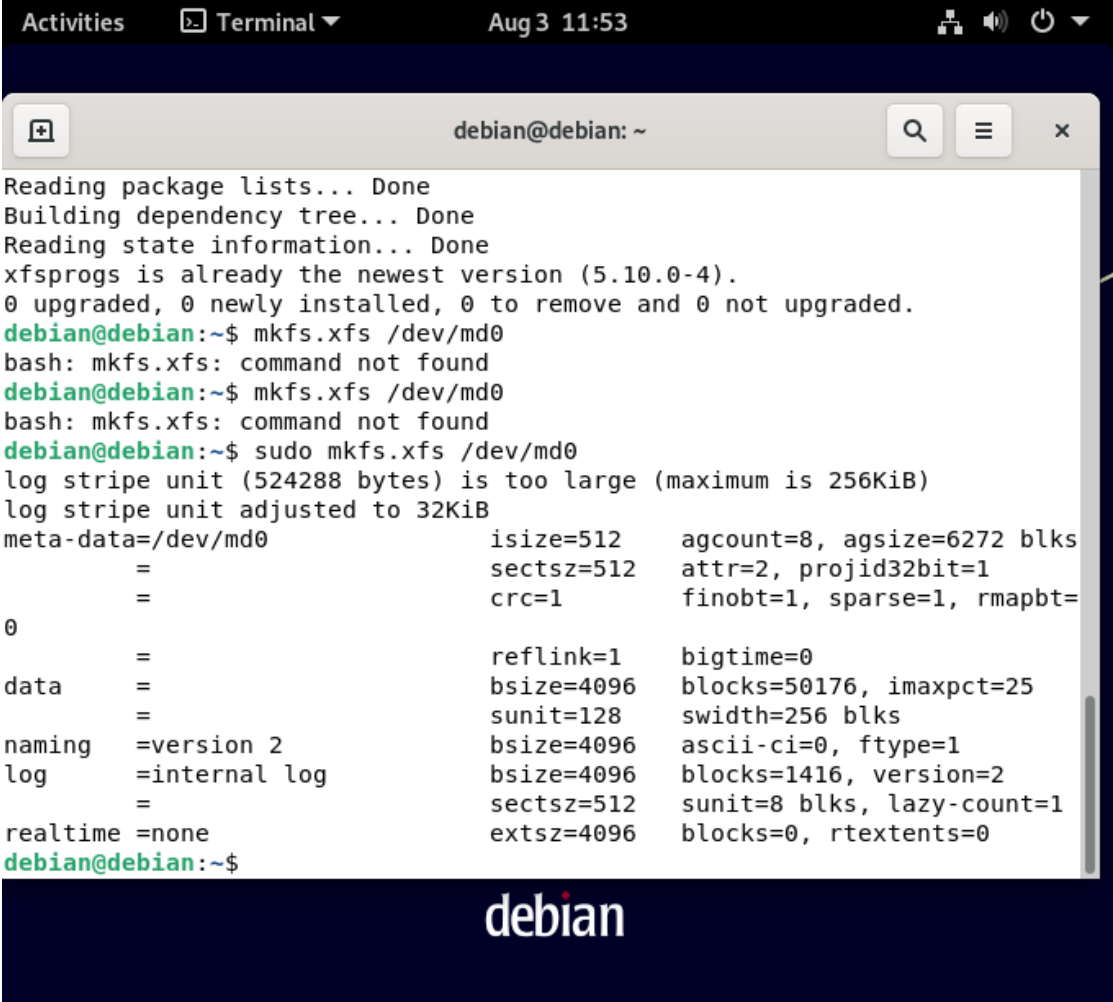


```
Activities  Terminal ▾  Aug 3 11:48  [Icons] [Volume] [Power] ▾

debian@debian: ~
mdadm: layout defaults to left-symmetric
mdadm: layout defaults to left-symmetric
mdadm: super1.x cannot open /dev/sda: Device or resource busy
mdadm: ddf: Cannot use /dev/sda: Device or resource busy
mdadm: Cannot use /dev/sda: It is busy
mdadm: cannot open /dev/sda: Device or resource busy
debian@debian:~$ sudo mdadm --create --verbose /dev/md0 --level=5 --raid-devices
=3 /dev/sdb /dev/sdc /dev/sdd
mdadm: layout defaults to left-symmetric
mdadm: layout defaults to left-symmetric
mdadm: chunk size defaults to 512K
mdadm: size set to 102400K
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
debian@debian:~$ sudo mdadm --add /dev/md0 /dev/sde
mdadm: added /dev/sde
debian@debian:~$ sudo apt-get install xfsprogs
[sudo] password for debian:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
xfsprogs is already the newest version (5.10.0-4).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
debian@debian:~$
```

Para criar o file system, usamos o seguinte comando: `sudo mkfs.xfs /dev/md0`

Para criar um ponto de montagem para o ficheiro system em “mnt/md0” usamos o seguinte comando: `sudo mkdir /mnt/md0`. Para a montagem, usamos o comando: `sudo mount /dev/md0 /mnt/md0`.



```
Activities  Terminal  Aug 3 11:53

debian@debian: ~
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
xfsprogs is already the newest version (5.10.0-4).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
debian@debian:~$ mkfs.xfs /dev/md0
bash: mkfs.xfs: command not found
debian@debian:~$ mkfs.xfs /dev/md0
bash: mkfs.xfs: command not found
debian@debian:~$ sudo mkfs.xfs /dev/md0
log stripe unit (524288 bytes) is too large (maximum is 256KiB)
log stripe unit adjusted to 32KiB
meta-data=/dev/md0          isize=512    agcount=8, agsize=6272 blks
                        =               sectsz=512   attr=2, projid32bit=1
                        =               crc=1        finobt=1, sparse=1, rmapbt=
0
                        =               reflink=1    bigtime=0
data          =             bsize=4096   blocks=50176, imaxpct=25
                        =             sunit=128   swidth=256 blks
naming        =version 2     bsize=4096   ascii-ci=0, ftype=1
log           =internal log  bsize=4096   blocks=1416, version=2
                        =             sectsz=512   sunit=8 blks, lazy-count=1
realtime      =none         extsz=4096   blocks=0, rtextents=0
debian@debian:~$
```

debian

Para garantirmos que inicia no instante 0 0, usamos o seguinte comando: `/dev/md0`
`/mnt/md0 xfs defaults 0 0` no código do ficheiro `fstab`



```

GNU nano 5.4 /etc/fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# systemd generates mount units based on this file, see systemd.mount(5).
# Please run 'systemctl daemon-reload' after making changes here.
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/sda1 during installation
UUID=cbf8d401-40bb-4eed-aeb9-4423d597bfd7 / ext4 errors>
# swap was on /dev/sda5 during installation
UUID=c034528f-6062-4650-8e4f-9cf703972e1b none swap sw >
/dev/sr0 /media/cdrom0 udf,iso9660 user,noauto 0 0

[ File '/etc/fstab' is unwritable ]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify
  
```

debian

- iii. Confirme o layout do seu array inspecionando os detalhes do array / dev / md0.

```

Activities  Terminal  Aug 3 12:04
debian@debian: ~

Version : 1.2
Creation Time : Wed Aug  3 11:28:45 2022
Raid Level : raid5
Array Size : 204800 (200.00 MiB 209.72 MB)
Used Dev Size : 102400 (100.00 MiB 104.86 MB)
Raid Devices : 3
Total Devices : 4
Persistence : Superblock is persistent

Update Time : Wed Aug  3 11:54:54 2022
State : clean
Active Devices : 3
Working Devices : 4
Failed Devices : 0
Spare Devices : 1

Layout : left-symmetric
Chunk Size : 512K

Consistency Policy : resync

Name : debian:0 (local to host debian)
UUID : 5faa6d3b:1cf5852a:249ce1c5:7c73edb4
Events : 20

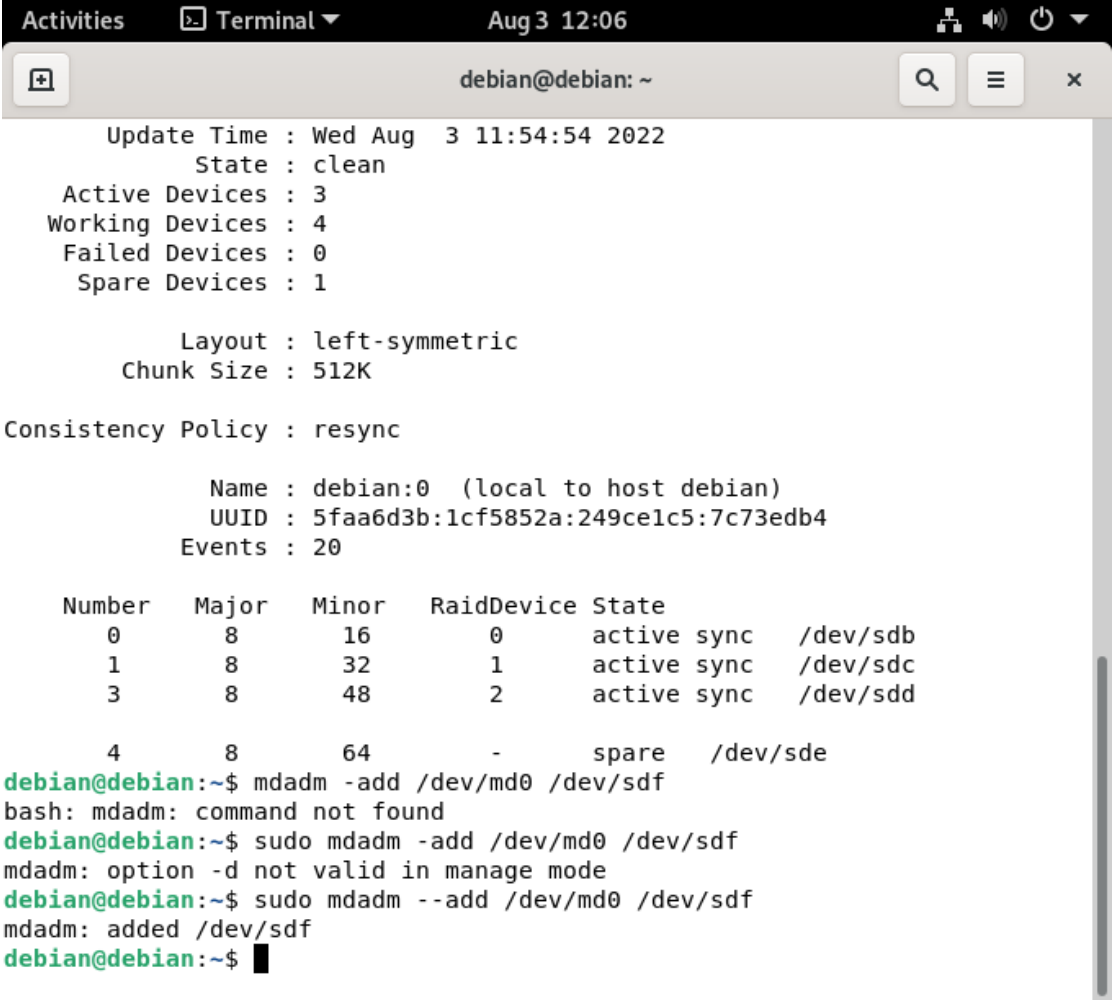
Number  Major  Minor  RaidDevice State
  0       8    16        0  active sync  /dev/sdb
  1       8    32        1  active sync  /dev/sdc
  3       8    48        2  active sync  /dev/sdd
    
```

Para confirmarmos o layout do array usamos o seguinte comando: `sudo mdadm -D /dev/md0`

iv. Adicione um quarto disco ativo ao array e redimensione o sistema de arquivos que você criou em /dev/md0.

Para adicionarmos o quarto disco, usamos o seguinte comando:

```
sudo mdadm --add /dev/md0 /dev/sdf
```



```

Activities  Terminal  Aug 3 12:06
debian@debian: ~
Update Time : Wed Aug  3 11:54:54 2022
State : clean
Active Devices : 3
Working Devices : 4
Failed Devices : 0
Spare Devices : 1

Layout : left-symmetric
Chunk Size : 512K

Consistency Policy : resync

Name : debian:0 (local to host debian)
UUID : 5faa6d3b:1cf5852a:249ce1c5:7c73edb4
Events : 20

Number   Major   Minor   RaidDevice State
  0         8       16         0   active sync  /dev/sdb
  1         8       32         1   active sync  /dev/sdc
  3         8       48         2   active sync  /dev/sdd

  4         8       64         -   spare        /dev/sde
debian@debian:~$ mdadm -add /dev/md0 /dev/sdf
bash: mdadm: command not found
debian@debian:~$ sudo mdadm -add /dev/md0 /dev/sdf
mdadm: option -d not valid in manage mode
debian@debian:~$ sudo mdadm --add /dev/md0 /dev/sdf
mdadm: added /dev/sdf
debian@debian:~$
  
```

Para aumentar o tamanho do array usamos o comando:

```
sudo mdadm --grow --raid-devices=4 /dev/md0
```

Para aumentar o tamanho do ficheiro system usamos o seguinte comando:

```
sudo xfs_growfs -d /mnt/md0
```

```

Activities  Terminal  Aug 3 12:08
debian@debian: ~

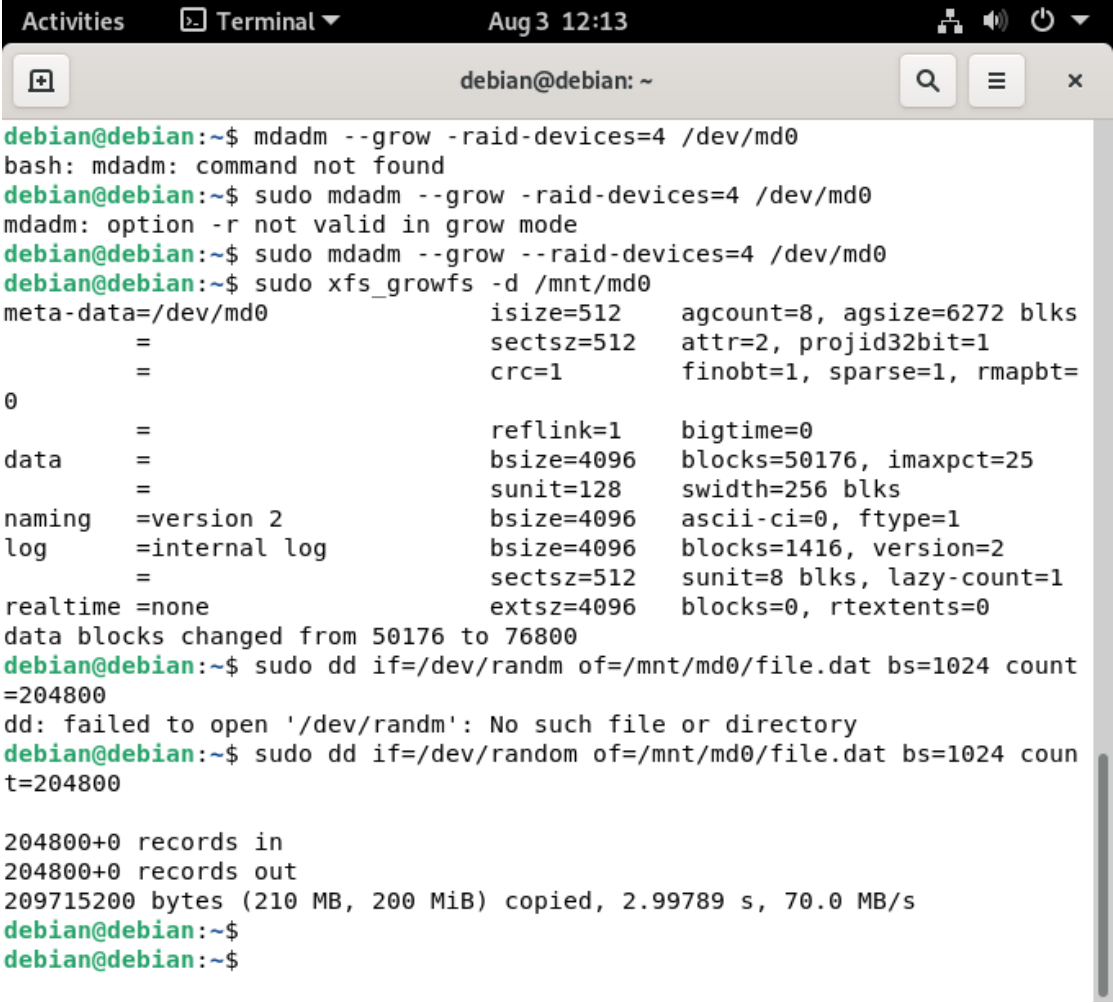
1      8      32      1      active sync  /dev/sdc
3      8      48      2      active sync  /dev/sdd

4      8      64      -      spare    /dev/sde
debian@debian:~$ mdadm -add /dev/md0 /dev/sdf
bash: mdadm: command not found
debian@debian:~$ sudo mdadm -add /dev/md0 /dev/sdf
mdadm: option -d not valid in manage mode
debian@debian:~$ sudo mdadm --add /dev/md0 /dev/sdf
mdadm: added /dev/sdf
debian@debian:~$ mdadm --grow -raid-devices=4 /dev/md0
bash: mdadm: command not found
debian@debian:~$ sudo mdadm --grow -raid-devices=4 /dev/md0
mdadm: option -r not valid in grow mode
debian@debian:~$ sudo mdadm --grow --raid-devices=4 /dev/md0
debian@debian:~$ sudo xfs_growfs -d /mnt/md0
meta-data=/dev/md0          isize=512    agcount=8, agsize=6272 blks
                =          sectsz=512    attr=2, projid32bit=1
                =          crc=1        finobt=1, sparse=1, rmapbt=
0
                =          reflink=1    bigtime=0
data        =          bsize=4096    blocks=50176, imaxpct=25
                =          sunit=128   swidth=256 blks
naming      =version 2          bsize=4096    ascii-ci=0, ftype=1
log         =internal log      bsize=4096    blocks=1416, version=2
                =          sectsz=512   sunit=8 blks, lazy-count=1
realtime    =none             extsz=4096    blocks=0, rtextents=0
data blocks changed from 50176 to 76800
debian@debian:~$
  
```

- v. Use uma combinação de dd e /dev/random para gerar o ficheiro /mnt/md0/file.dat com 200 MB de tamanho

Para gerar o ficheiro, usamos o seguinte comando:

```
sudo dd if=/dev/random of=/mnt/md0/faile.dat bs=1024  
count=204800
```



```
Activities Terminal Aug 3 12:13  
debian@debian: ~  
debian@debian:~$ mdadm --grow -raid-devices=4 /dev/md0  
bash: mdadm: command not found  
debian@debian:~$ sudo mdadm --grow -raid-devices=4 /dev/md0  
mdadm: option -r not valid in grow mode  
debian@debian:~$ sudo mdadm --grow --raid-devices=4 /dev/md0  
debian@debian:~$ sudo xfs_growfs -d /mnt/md0  
meta-data=/dev/md0          isize=512    agcount=8, agsize=6272 blks  
                =          sectsz=512   attr=2,   projid32bit=1  
                =          crc=1        finobt=1, sparse=1, rmapbt=  
0  
                =          reflink=1    bigtime=0  
data          =          bsize=4096    blocks=50176, imaxpct=25  
                =          sunit=128   swidth=256 blks  
naming        =version 2          bsize=4096   ascii-ci=0, ftype=1  
log           =internal log      bsize=4096   blocks=1416, version=2  
                =          sectsz=512   sunit=8 blks, lazy-count=1  
realtime      =none             extsz=4096    blocks=0, rtextents=0  
data blocks changed from 50176 to 76800  
debian@debian:~$ sudo dd if=/dev/randm of=/mnt/md0/file.dat bs=1024 count  
=204800  
dd: failed to open '/dev/randm': No such file or directory  
debian@debian:~$ sudo dd if=/dev/random of=/mnt/md0/file.dat bs=1024 coun  
t=204800  
  
204800+0 records in  
204800+0 records out  
209715200 bytes (210 MB, 200 MiB) copied, 2.99789 s, 70.0 MB/s  
debian@debian:~$  
debian@debian:~$
```

vi. Use mdadm para causar uma falha num dos discos ativos e remova-o do array.

Para causar uma falha num dos discos ativos, usamos o seguinte comando:

```
sudo mdadm /dev/md0 -f /dev/sdc
```

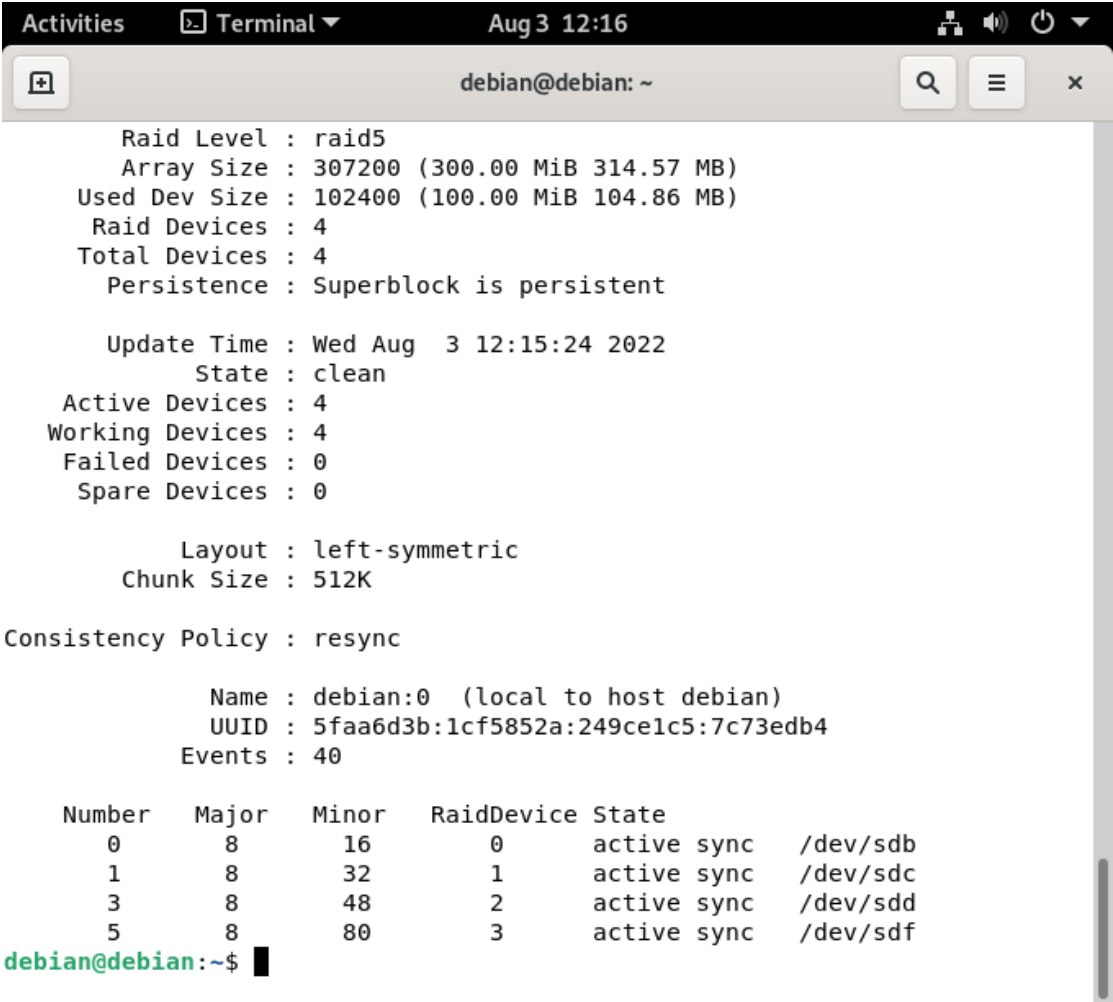
Para remover o disco, usamos o seguinte comando:

```
sudo mdadm /dev/md0 -r /dev/sdc
```

vii. Confirme se um dos discos hot-spare tomou seu lugar, inspecionando os detalhes do array.

É usado o seguinte comando:

```
sudo mdadm -D /dev/md0
```



```

Activities  Terminal  Aug 3 12:16
debian@debian: ~
Raid Level : raid5
Array Size : 307200 (300.00 MiB 314.57 MB)
Used Dev Size : 102400 (100.00 MiB 104.86 MB)
Raid Devices : 4
Total Devices : 4
Persistence : Superblock is persistent

Update Time : Wed Aug 3 12:15:24 2022
State : clean
Active Devices : 4
Working Devices : 4
Failed Devices : 0
Spare Devices : 0

Layout : left-symmetric
Chunk Size : 512K

Consistency Policy : resync

Name : debian:0 (local to host debian)
UUID : 5faa6d3b:1cf5852a:249ce1c5:7c73edb4
Events : 40

Number Major Minor RaidDevice State
0 8 16 0 active sync /dev/sdb
1 8 32 1 active sync /dev/sdc
3 8 48 2 active sync /dev/sdd
5 8 80 3 active sync /dev/sdf
debian@debian:~$
  
```

5. Pesquise e descreva:

- i. As funcionalidades básicas do Volume Snapshot Service (VSS) no Windows

O VSS é uma tecnologia que permite fazer backups enquanto as aplicações que estão a ser armazenadas continuam a ler os dados. Permite ainda realizar pequenas cópias instantâneas através do snapshot.

- ii. Os motivos pelos quais o VSS não é considerado uma tecnologia de backup

O VSS não é considerado uma tecnologia de backup porque não contém a cópia de todos os dados mas apenas os dados que foram alterados até à imagem.